

Yi Hu

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PERSONAL SUMMARY

PhD researcher specialized in **embodied AI** and **surgical robotics**. Developed **Vision-Language-Action (VLA) model** and **Reinforcement Learning (RL)/Imitation Learning (IL)** frameworks for Franka and da Vinci surgical robots. Strong background in SLAM, kinematics/dynamics, and control theory.

EDUCATION

PhD in Biomedical Engineer <i>University of Alberta</i>	Jan. 2021 – Exp. Apr. 2026 <i>Edmonton, Canada</i>
MS in Control Theory and Control Engineering <i>Sichuan University</i>	Sept. 2017 – Jun. 2020 <i>Chengdu, China</i>
BS in Process Equipment and Control Engineering <i>Hefei University of Technology</i>	Sept. 2013 – Jun. 2017 <i>Hefei, China</i>

PROJECT EXPERIENCE

Human-Centered Autonomy Lab <i>Graduate Researcher, University of Alberta</i>	Jan. 2024 – Present <i>Edmonton, Canada</i>
- Designed a VLA policy integrating 3D point cloud reconstruction (PointNet) and VLM feature projection, fused via a Mixture-of-Experts (MoE) action head to condition robot joints; trained on the LIBERO-Spatial environment (10 reasoning tasks with 500+ demonstrations), achieving >96% success rate. - Built an end-to-end VLA training and deployment pipeline: collected and post-trained a Pi0.5 model on >300GB multimodal robot data across 5 manipulation tasks with 750+ demonstrations, and deployed real-time inference on a real-world Franka robot, achieving >95% success rate. - Engineered a real-time motion policy for dynamic fast manipulation by replacing multi-step diffusion sampling with a single-step diffusion model; trained on 100 demonstrations, achieving 7–10 inference latency reduction and validated on a real Franka robot ball-catching task. - Developed a RL framework that transfers from large-scale human activity datasets (32+ tool-use tasks) to robotic surgical subtasks, enabling precise patient-side manipulator and endoscopic camera control with 13.5 improved sample efficiency; validated via sim2real transfer from PyBullet to a real-world da Vinci surgical robot.	
Telerobotic and Biorobotic Systems Group <i>Graduate Researcher, University of Alberta</i>	Jan. 2021 – Dec. 2023 <i>Edmonton, Canada</i>
- Built an end-to-end autonomous ultrasound scanning system on a Franka robot, integrating classic control using an Axia80-M20 force-torque sensor with UNet-based deep learning image processing via DVI2USB video capture, achieving >80% accuracy at >10 Hz closed-loop control. - Developed an autonomous endoscopic camera control system and patient side manipulator control system for the da Vinci surgical robot operating on real-world noisy surgical data, with automatic data processing of unsafe or failed actions and a safety-critical, collision-aware motion planning framework. - Developed a real-time teleoperation ultrasound scanning pipeline that transformed free-hand movements into 6-DoF robot control; Integrated a lightweight gesture classifier trigger discrete robot actions.	
Perception, Motion Control and Intelligent Robot Innovation Lab <i>Graduate Researcher, Sichuan University</i>	Sept. 2017 – Dec. 2020 <i>Chengdu, China</i>
- Implemented a real-time event-camera SLAM pipeline for mobile robot navigation with scene understanding, enabling 6-DoF motion tracking and 3D reconstruction through optimization-based mapping and keyframe back-end refinement. Built a physics-based modeling in Unity for mobile robot navigation. - Designed robust nonlinear control algorithms for joint-space control, accounting for actuator saturation and incorporated disturbance observers, with system stability guaranteed through Lyapunov analysis. - Derived kinematic and dynamic models for a wheel mobile robot, enabling motion planning, control design, and trajectory tracking.	

SKILLS

Programming: Python, MATLAB/Simulink, C++
Robotics: ROS, ROS2, MuJoCo, PyBullet, Isaac Gym, Gazebo, RViz
ML & AI: PyTorch, TensorFlow, Diffusion Models, Transformers, RL, IL
Systems & Tools: Linux, Git, CUDA, SolidWorks, COMSOL, AutoCAD